



Data Visualization Techniques

Exercise 3 Report

Prof: Daniel Vitaver

Group 7

Gavriel Kirichenko (101119609)

Bikash Giri (101575097)

Chia-Wei Chang (101570243)

Hsi-Teng Liu (101576074)

Abdul-Rasaq Omisesan (101571156)

Callum Arul (101585383)

Friba Hussainyar (101591222)

Diparshan Bhattarai (101494737)

Sergio Alejandro Medrano Díaz (101446299)

Objective

This study uses Tableau's analytical models and advanced data visualizations to examine global cybersecurity concerns. The core goals were to:

- Analyze the relationship between industries and attack types.
- Quantify user and financial impacts by sector and geography.
- Group economies based on exposure to cyber threats.
- Find which defense strategies performed best against different attack types
- Support strategic decision-making through KPI scorecards.

Tableau features employed include distribution histograms, autocorrelation matrices, Pareto charts, clustering, regression, and interactive dashboards with filters and Python-powered calculated fields.

Operational Dashboard

Autocorrelation Matrix – Attack Type Relationships

The autocorrelation matrix uncovers statistical relationships between different cyberattack types by measuring how frequently they co-occur within the same incidents or environments. The correlation coefficients range from **0.1832** (very weak) to **0.7681** (strong), providing insight into which threats tend to appear together, and by extension, which ones may be targeting similar vulnerabilities.

Key Attack Pair Relationships

- **Malware & Ransomware (0.7681):**
This is the strongest correlation in the dataset. Ransomware attacks almost always involve malware as the delivery mechanism as it's the trojan horse that slips through a vulnerability before locking down systems. This high correlation reflects a common real-world pattern: ransomware is rarely the first step; it piggybacks off malware infections.
- **Phishing & Man-in-the-Middle (0.6446):**
These two attack types are often part of social engineering and interception strategies. Phishing lures users into divulging credentials or clicking malicious links, while man-in-the-middle attacks exploit those compromised channels. Their tight relationship suggests that once an attacker gains initial access (via phishing), they often escalate using network-level manipulation.
- **DDoS & Malware (0.6059):**
This pair might be overlooked at first, but the correlation suggests a coordinated

strategy: malware can be used to create botnets that power DDoS attacks. The implication is clear — if a system is infected with malware, it could be contributing to a much larger infrastructure of distributed attacks.

Patterns & Implications

Interestingly, all attack types show some level of interconnectedness, even SQL injection (albeit weak at **0.1832**). This hints at the increasingly complex and layered nature of modern cyberattacks. Rarely are systems targeted by just one type of threat. Instead, attackers leverage multiple tools in tandem to maximize disruption or extract greater value.

Strategic Takeaways

- **Defense Can't Be One-Dimensional:**
Organizations need to stop treating threats in isolation. A response plan focused solely on ransomware, for example, might fail if it ignores the malware stage that comes before it.
- **Prioritize Entry Point Protection:**
Phishing and malware are often the first signs of deeper, more sophisticated attacks. Catching and neutralizing them early can prevent downstream damage from ransomware, DDoS, or MITM attacks.
- **Invest in Behavioral Analytics:**
Given the correlations, detecting patterns of combined activity (e.g., malware plus DDoS traffic spikes) can serve as early warning signs. Solutions like SIEM or XDR platforms with autocorrelation logic should be part of the cybersecurity stack.

In short, this matrix doesn't just show us which attacks are statistically linked, it reveals how attackers think. And if we want to stay ahead, our defenses need to reflect that same level of coordination.

Pareto Analysis – Financial Loss Concentration

This Pareto chart sheds light on one of the most critical insights in cybersecurity economics: **a small number of threat vectors are responsible for the vast majority of financial damage.**

By plotting the cumulative financial loss against the cumulative count of unique attack combinations (grouped by source, industry, and geography), the chart visually confirms the classic **80/20 rule** — roughly **80% of global cyber-related financial losses can be traced back to just 20% of attack sources.**

What the Chart Tells Us

- The sharply curved **red cumulative line** illustrates how rapidly losses accumulate from just a few attack combinations.
- The **vertical black line** at the 20% mark acts as a threshold. Everything to the left of it represents the most devastating combinations in terms of economic impact.
- The **dot on the curve** pinpoints the critical inflection point: where focusing resources yields the highest marginal return in loss prevention.
- The **green-to-red color gradient** underlines risk concentration, with the earliest segments representing the most harmful combinations.

Why This Matters

This visualization goes beyond surface-level numbers, it highlights the **disproportionate impact** of a few specific threat vectors. It's not just that a handful of attacks are common; it's that they cause **catastrophic damage** when they occur.

For example, the combination of phishing in the finance sector in North America, or ransomware attacks in healthcare across Europe, might fall within this top 20%. These aren't necessarily the most frequent threats, they're the ones that **do the most financial harm**.

Strategic Recommendations

- **Target the "Vital Few":**
Cybersecurity teams and policymakers should prioritize the top 20% of high-risk threat combinations. This doesn't mean ignoring the rest, but rather **allocating disproportionate resources, tools, personnel, and budget, to the areas that matter most**.
- **Inform Budgeting Decisions:**
When budgets are limited (as they often are), this chart can guide where to spend. Focusing on this slice could result in **outsized reductions in loss**, potentially saving hundreds of millions.
- **Refine Mitigation Strategy by Sector and Geography:**
Because the underlying data includes attack type, industry, and country, this analysis can be broken down further to drive **region-specific** or **industry-specific policies**. A generic "one size fits all" response won't cut it. Precision targeting is both more efficient and more effective.

What If? – Cybersecurity Investment vs. Global Ranking

This bar chart visualizes the possible impact of increased cybersecurity investment on a country's ability to manage threats effectively, as measured by average global ranking. It's a forward-thinking analysis posing the question: *What happens when countries commit more resources to digital defense?*

Key Observations

- **France** stands out as the clear leader, with over **\$2 billion** in cybersecurity investment and a **top-tier global rank of 13.09**. It's not just spending more, it's seeing results. The green upward arrow emphasizes this positive correlation: money well spent leads to improved defense posture.
- **Russia** and **India**, by contrast, are investing significantly less, around **\$630 million** and **\$560 million**, respectively, and their rankings reflect this with noticeable underperformance. Both are flagged with **orange downward arrows**, signaling regression or stagnation despite growing threats.
- **Brazil**, with the **lowest investment (\$180 million)**, ranks the worst of the group at **27.77**, and is double flagged with decline indicators. Its case underlines the risk of underinvestment in a rapidly evolving threat landscape.
- **Australia** holds a middle-ground position, moderate investment and an average ranking, acting as a useful benchmark between the extremes.

The Bigger Picture

The visual makes a strong, intuitive case: **countries that invest more in cybersecurity tend to outperform their peers in threat mitigation and preparedness**. The average global rank (likely a composite measure involving response time, damage control, and infrastructure resilience) serves as a tangible output of a nation's cyber capabilities.

This is especially relevant for **emerging economies**, where investment gaps can result in disproportionate exposure to threats, not just due to outdated infrastructure, but also due to lack of training, public-private coordination, and national cyber frameworks.

Recommendations

- **Policy Advocacy for Emerging Markets:**
This chart should serve as a wake-up call. Countries like Brazil and India need to **scale up funding, personnel training, and strategic partnerships** to catch up with global leaders like France.
- **International Benchmarking:**
Cybersecurity strategies shouldn't be built in isolation. Governments can use this data

to **benchmark their progress** against peers and identify where increased funding would yield the highest returns.

- **Public-Private Investment Models:**

For countries unable to match public sector funding, this insight can drive **joint investment models**, encouraging private sector stakeholders to co-invest in national cybersecurity resilience.

Clustering Analysis – GDP vs. Financial Loss

This scatter plot, enhanced with **K-means clustering**, groups countries based on two variables: **GDP** (in trillions USD) and **financial loss from cyberattacks** (in millions). Each cluster reveals a distinct economic-cybersecurity profile, helping to explain why different nations experience vastly different impacts, even when facing similar threats.

Cluster Profiles

- **Cluster 1 (Blue): Low GDP, Low Loss**

These are developing nations with limited digital infrastructure and minimal exposure to large-scale attacks. While their losses appear low, it's often due to underreporting or lack of high-value targets, not necessarily strong cybersecurity.

- **Cluster 2 (Orange): Low GDP, High Loss**

Perhaps the most alarming group. These countries are being **targeted disproportionately relative to their economic strength**, often because they lack resources to defend themselves. They're **high-risk, low-preparedness zones** that attract opportunistic attackers.

- **Cluster 3 (Red): Medium GDP, Medium Loss**

A transitional group, often countries that are expanding their digital economies but haven't yet fully matured their cyber defenses. They're **on the radar**, and their losses are climbing with economic activity.

- **Cluster 4 (Green): High GDP, High Loss**

Wealthy, digitally advanced nations. Unsurprisingly, they bear the brunt of financial damage. These countries are **high-value targets** for cybercriminals and nation-state attackers due to their digital dependency and potential for large payouts.

Key Insight

There's a **clear upward trend**: as GDP increases, so does the financial fallout from cyber threats. Attackers follow the money, and the infrastructure. This correlation underscores the **economic motivation** behind most cyberattacks and exposes the vulnerability that comes with modernization.

Strategic Implications

- **For Developing Nations (Clusters 1 & 2):**
 - **Cluster 2 is a priority.** These countries face high damage with little means of defense. International aid, knowledge transfer, and subsidized security frameworks could be life-saving, literally and economically.
 - Build foundational cybersecurity capacity: endpoint protection, incident response training, and basic regulatory oversight.
- **For Transitioning Economies (Cluster 3):**
 - Introduce **national cybersecurity strategies** before exposure escalates.
 - Foster partnerships between government, private sector, and international cybersecurity organizations to prepare for future threats.
- **For Wealthy Nations (Cluster 4):**
 - **Invest in resilience.** Being a prime target is unavoidable, but damage isn't. Focus on AI-driven threat detection, regulatory compliance, and public-private intelligence sharing.
 - Consider helping secure global supply chains, attacks often reach rich nations through compromised vendors in poorer ones.

Executive Scorecard – KPIs for Monitoring & Strategy

The executive scorecard presents a comparative view of key cybersecurity indicators, enabling decision-makers to assess year-over-year changes and effectiveness by region, industry, and defense mechanism.

The executive scorecard includes global filters for **Country**, **Target Industry**, **Current Year**, and **Previous Year**. These filters allow users to explore trends, compare defense effectiveness, and assess threat impact across different years and industry contexts.

KPI 1: Top & Bottom Performing Cyber Defense Mechanisms (2023 → 2024)

This dual-panel scorecard highlights the cyber defense mechanisms that saw the most significant changes in performance between 2023 and 2024, as measured by their **impact on average financial loss per incident**. The chart provides a clear before-and-after snapshot, helping decision-makers quickly assess where defenses are working, and where they're falling short.

Top Performing Defenses

1. Firewall

- **2024 Loss:** \$47.22M
- **Improvement:** ▼ \$6.08M
- Firewalls showed the **strongest improvement over the last year**, reducing average incident losses by over \$6 million. This suggests improvements in rule sets, detection logic, or perhaps better integration with modern network architectures.

2. Encryption

- **2024 Loss:** \$45.54M
- **Improvement:** ▼ \$1.73M
- Encryption continues to provide reliable baseline protection, with moderate gains likely linked to broader adoption of zero-trust and end-to-end encrypted channels.

3. Antivirus

- **2024 Loss:** \$56.50M
- **Decline:** ▲ \$1.87M
- Despite appearing in the “Top” section, Antivirus actually posted a **negative delta**, underscoring the limitations of signature-based protection in the face of modern, polymorphic threats. Its position here reflects high baseline usage rather than recent effectiveness.

Least Performing Defenses

1. VPN

- **2024 Loss:** \$54.46M
- **Decline:** ▲ \$6.41M
- VPNs took the biggest hit. With the rise of sophisticated tunneling attacks and compromised endpoints, traditional VPNs appear to be increasingly outmatched by attackers. This raises urgent questions about whether organizations have transitioned to more secure frameworks like ZTNA (Zero Trust Network Access).

2. AI-based Detection

- **2024 Loss:** \$53.75M
- **Decline:** ▲ \$3.70M
- Despite being a buzzword in cybersecurity, AI-based detection tools underperformed this cycle. The decline may reflect either model drift (where algorithms degrade without retraining) or gaps in threat intelligence pipelines.

3. Antivirus

- **2024 Loss:** \$56.50M
- **Decline:** ▲ \$1.87M
- Appearing in both tables, Antivirus illustrates an important nuance: while still widely used, its **relative performance** has dropped, especially in fast-moving threat environments.

Takeaway

This KPI offers a **clear performance audit of cyber defenses**. By tracking which mechanisms are losing ground and which are improving, organizations can better allocate resources, retire ineffective tools, and double down on high-performing ones. It's not just about spending more, it's about spending **smarter**.

KPI 2: Threat Score by Attack Type (2023 → 2024)

This horizontal bar chart ranks cyberattack types using a **composite threat score**, calculated via a custom Python model. The **threat score** is a weighted composite metric designed to quantify the severity of each attack type. It combines **normalized financial loss (50%)**, **number of users affected (30%)**, and **incident count (20%)**, placing the most emphasis on economic impact.

- **Financial loss** carries the most weight (50%), reflecting the real-world cost and organizational harm caused by an attack. This prioritizes threats that are **economically devastating**, even if they're less frequent.
- **User impact** (30%) captures the **human cost** of breaches, especially relevant for sectors like healthcare, finance, and education where sensitive data is involved.
- **Incident frequency** (20%) ensures that **common but lower-damage attacks** (like phishing or SQL injection) aren't ignored.

By normalizing each input, the score allows for **fair comparison across diverse attack types**, and the weighted formula ensures that the final score reflects a **strategic blend of frequency and severity**, making it ideal for prioritizing defensive action.

Rising Threats

- **Malware**
The most alarming figure on the board, malware tops the chart with a threat score over 80, the highest among all categories. Its score **increased** from 2023, suggesting either wider spread, greater impact, or both. Malware continues to evolve, often serving as the foundation for more complex, multi-stage attacks like ransomware or data exfiltration.
- **Ransomware, Phishing, and Man-in-the-Middle**
All three saw moderate score increases, underscoring their **continued relevance and growing sophistication**. These attacks are frequently used in combination, with phishing acting as an entry vector and ransomware or MITM techniques following soon after.

Declining Threats

- **DDoS (Distributed Denial of Service)**
DDoS dropped in threat score, possibly reflecting better cloud-based mitigation tools and proactive detection systems. While still disruptive, these attacks are easier to deflect today than they were a few years ago.
- **SQL Injection**
The decline here may reflect improved input sanitization, developer training, and the widespread adoption of modern frameworks that inherently protect against basic injection flaws. It's a positive signal, but not a reason to become complacent.

Visual Cues

- **Red bars** represent threat types with **rising scores**.
- **Green bars** represent those with **declining scores**.
- The **black arrows** beside each bar indicate direction of change, helping users instantly identify year-over-year momentum.

Strategic Takeaways

- **Malware must remain a top priority.** Its versatility and role in other major attacks make it a foundational threat that can't be ignored.
- **Ransomware and phishing** remain persistently dangerous and should be addressed with targeted training, endpoint protection, and email filtering systems.
- **Legacy threats like SQL injection** may be declining but still represent low-effort, high-reward opportunities for attackers in poorly maintained systems.

This KPI enables security leaders to track shifting threat dynamics and **recalibrate their defense strategy** accordingly, focusing not just on what's common, but on what's **growing** in risk.

KPI 3: Average Financial Loss Per Incident (2023 → 2024)

This bar chart highlights the **average financial damage per incident** for each major cyberattack type, comparing values between 2023 and 2024. While overall frequency matters, this KPI zooms in on **how costly each incident is when it does occur**, offering a clearer lens on high-impact threats.

Key Findings

- **Malware** once again tops the list with an average loss of **\$55.95M per incident**, rising from the previous year. Its escalating cost underscores its evolving sophistication, often used as the launchpad for deeper intrusions like ransomware or data exfiltration.
- **Man-in-the-Middle, Ransomware, and Phishing** all hover around the **\$51M–52M range**, each showing **year-over-year increases**. This suggests that even "classic" attack types are becoming more expensive to recover from, either due to deeper data compromise, longer system downtime, or steeper regulatory penalties.
- **DDoS**, while still costly at **\$52.43M**, saw a **decline** in average loss. This may reflect improved preparedness through cloud-based mitigation or the declining financial impact of bandwidth-based disruptions alone.
- **SQL Injection** comes in lowest at **\$45.13M**, and also saw a **year-over-year decrease**, possibly due to better development practices, code review automation, and framework protections.

Design Details

- Arrows indicate trend direction:
 - ▲ Increase in loss from 2023 to 2024
 - ▼ Decrease in loss
- **Color-coded bars** differentiate performance trends:
 - Orange = rising average cost
 - Blue = declining or neutral performance

Strategic Takeaway

Tracking **cost per incident** is critical for budgeting and prioritization. An attack might not happen often, but when it does, the damage can be severe. This KPI allows organizations to **focus investments on minimizing high-cost incidents**, even if their occurrence rate is low. It also helps justify investment in **proactive threat detection and mitigation**, especially for attacks with rising cost curves like malware and phishing.

KPI 4: Cybersecurity Incident Frequency Over Time (2015–2024)

This line chart tracks the **total number of reported cybersecurity incidents** annually over a ten-year span, offering a macro-level view of threat volume trends. The graph includes **year-over-year change labels** and a **dashed linear trendline**, allowing viewers to distinguish short-term fluctuations from long-term patterns.

Key Patterns

- **Peak Activity** occurred in **2017** with **319 incidents**, marking a sharp rise of 34 incidents from the previous year, potentially tied to global malware outbreaks like WannaCry and NotPetya.
- **Significant drop** in 2019 (**-47 incidents**) may reflect temporary threat suppression, improved defensive posture, or reporting lags in certain regions.
- Incident volume rebounded in **2020** with a **52-incident increase**, aligning with pandemic-era surges in digital activity and remote work vulnerabilities.
- **Stabilization phase** (2021–2024): Incident counts hovered around 299–318 per year, suggesting a plateau in reporting or threat frequency, though this doesn't necessarily reflect severity.

Visual Features

- Each point is annotated with:
 - The **incident count** for the year.
 - The **delta from the previous year**, with red and green markers denoting increase or decrease.
- The **dashed regression line** provides context for overall trend movement, which appears **mostly flat**, indicating **no clear linear growth or decline**.

Dashboard Integration

This KPI connects to the rest of the dashboard via **global filters** for:

- Country
- Target Industry
- Current Year
- Previous Year

This allows users to **narrow the timeline to specific regions or sectors**, revealing spikes or drops tied to particular geopolitical events or policy changes.

Strategic Takeaway

While the total number of incidents has remained relatively stable in recent years, this **should not be mistaken for risk reduction**. The rising **cost per incident** (as seen in other KPIs) suggests that attacks are becoming **more targeted and impactful**, not necessarily more frequent. Monitoring this metric in tandem with loss and threat type KPIs provides a **fuller picture of evolving cyber risk**.

Root Cause Analysis & Strategic Recommendations

A deeper look into the data reveals that the cybersecurity challenges we face are **not just technical, they're systemic**. Patterns of risk concentration, infrastructure vulnerability, and uneven global preparedness point to several core issues that require both strategic foresight and cross-sector cooperation.

Key Challenges

1. **Sector-Specific Risk Concentration**

Industries like **finance, healthcare, and education** continue to experience a disproportionate number of high-impact incidents. These sectors house large volumes of sensitive data and often rely on legacy systems, making them **prime targets** for attackers.

2. **Emerging Economies in the Crosshairs**

Countries with lower GDPs are increasingly facing **complex, high-damage attacks**, despite having fewer resources to defend themselves. This trend reflects an aggressive shift in attacker focus toward **soft targets with limited protection**.

3. **Overlapping Threat Vectors**

Modern cyberattacks are rarely isolated. A single incident may involve malware, phishing, and man-in-the-middle techniques working in tandem. This complexity **complicates detection and mitigation**, often overwhelming siloed or outdated security systems.

Root Drivers

- **Outdated Infrastructure:** Public and private institutions still operate on legacy systems vulnerable to basic exploits.
- **Training Gaps:** Lack of staff awareness and security hygiene continues to be a weak link, especially in user-facing roles.
- **Regulatory Inconsistencies:** Some countries lack enforceable data protection laws or breach reporting mandates, leading to underreporting and unaccountability.
- **Economic Incentives for Attackers:** As revealed in the GDP vs. Loss clustering, **wealthier nations attract more attacks**, but emerging economies suffer more when targeted due to inadequate recovery capacity.

Strategic Recommendations

• **Adopt Multi-Layered, Integrated Security Frameworks**

Defense needs to span beyond firewalls and antivirus. Integration of endpoint protection, behavioral analytics, intrusion detection, and zero-trust architectures is key.

- **Invest in Early Detection**

The earlier an attack is identified, the cheaper and less damaging it is to contain.

Prioritize detection mechanisms for **high-risk vectors** like phishing, DDoS, and malware.

- **Focus on the "Vital Few"**

Leverage findings from the Pareto analysis: a small group of attack sources causes most of the damage. Concentrating budget and personnel on these few can yield **outsized returns**.

- **Boost Global Cyber Resilience**

Provide **policy support, aid, and knowledge transfer** to help emerging economies build fundamental cybersecurity infrastructure. Global security is only as strong as its weakest node.

- **Customize Defenses by Industry**

One-size-fits-all security doesn't work. For example, telecom providers may need **anti-phishing simulations and DMARC policies**, while hospitals may prioritize ransomware backups and access control.

This analysis shows that cybersecurity isn't just a technical issue. It's a global problem that's deeply connected to how organizations, industries, and countries operate. Tackling the root causes now isn't just about preventing future attacks — it's about building smarter, more resilient digital systems for everyone.